

Vedant Palit

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EDUCATION

Indian Institute of Technology (IIT) Kharagpur, India

12/2021 – 05/2026

B.Tech in Industrial Engineering and M.Tech in Financial Engineering

8.74/10

Relevant Coursework: Linear Algebra, Numerical and Complex Analysis; Statistical Learning with Applications; Regression Analysis and Time Series Models; Scalable Data Mining

PUBLICATIONS

- When Forgetting is not Catastrophic: On the Mechanics of Spurious Forgetting - **Vedant Palit**, Florent Draye, Nicolas Zucchet, Zhijing Jin, Bernhard Schölkopf.
- How Do Linear Probes Emerge? A Circuit-Tracing Framework with Concept-Targeted Attribution - **Vedant Palit**, Florent Draye, Terry Jingchen Zhang, Bernhard Schölkopf, Zhijing Jin (*EMNLP 2026 Main Track*).
- Simulating Eutopia: Revisiting Long-term Fairness with Outcomes, Performativity, and Dynamics - **Vedant Palit**, Udvass Das, Brahim Driss, Debabrota Basu (*EWAF 2026 Extended Abstract - Spotlight, ICML 2026 AI4LAW Workshop*).
- CLT-Forge: A Scalable Library for Cross-Layer Transcoders and Attribution Graphs - Florent Draye, **Vedant Palit**, Abir Harrasse, Tung-Yu Wu, Jiarui Liu, Punya Syon Pandey, Roderick Wu, Chih-Hao Hsu, Terry Jingchen Zhang, Zhijing Jin, Bernhard Schölkopf (*EMNLP 2026 Demo Track*).
- Towards Vision-Language Mechanistic Interpretability: A Causal Tracing Tool for BLIP - **Vedant Palit**, Rohan Pandey, Aryaman Arora, Paul Pu Liang (*ICCV 2023 CLVL Workshop*).
- Forgotten Polygons: Multimodal Large Language Models are Shape-Blind - William Rudman*, Michal Golovanevsky*, Amir Bar, **Vedant Palit**, Yann LeCun, Carsten Eickhoff, Ritambhara Singh (*ACL 2025 Findings Track*).
- What Do VLMs NOTICE? A Mechanistic Interpretability Pipeline for Gaussian-Noise-free Text-Image Corruption and Evaluation - Michal Golovanevsky*, William Rudman*, **Vedant Palit**, Ritambhara Singh, Carsten Eickhoff (*NAACL 2025*).
- WellDunn: On the Robustness and Explainability of Language Models and Large Language Models in Identifying Wellness Dimensions - Seyedali Mohammadi, Edward Raff, Jinendra Malekar, **Vedant Palit**, Francis Ferraro, Manas Gaur (*EMNLP 2024 Blackbox NLP Workshop*).
- Knowledge Graph Guided Semantic Evaluation of Language Models For User Trust - Kaushik Roy, Tarun Garg, **Vedant Palit** (*IEEE CAI 2023*).
- Exploring The Potential of Large Language Models for Assisting with Mental Health Diagnostic Assessments - Kaushik Roy, Harshul Surana, Darssan Eswaramoorthi, Yuxin Zi, **Vedant Palit**, Ritvik Garimella, Amit Sheth (*ACM Transactions on Computing for Healthcare*).

RESEARCH EXPERIENCE

Pre-PhD Researcher | Empirical Inference Department | MPI-IS

Tübingen, Germany

Advised by Prof. Bernhard Schölkopf

Jul 2026 – Present

- Conceived and led a study of the mechanics of spurious forgetting, building a minimal associative-memory model that reproduces the collapse, recovery and erosion of old facts during fine-tuning.
- Showed that a shared, reversible shift hides old facts while fact-specific changes erode them, and validated this in a Transformer trained on synthetic data and in a pretrained language model.
- Studying whether sequence models trained only on action sequences build causally used world-state representations, and how the training objective and memory architecture determine which state is kept and whether it persists.

Interpretability Lead | Jinesis Lab, University of Toronto

Hybrid

Advised by Dr. Zhijing Jin

Jan 2026 – Present

- Introduced Concept-Targeted Attribution, which builds attribution graphs toward linear probe directions; showed that concept-encoding and token-generation circuits are causally separable and that graph topology predicts probe accuracy across Gemma-2-2B and Llama-3.2-1B.
- Devised a compression and quantization framework for the efficient saving and loading of activations from GPT-2 and Llama-1B, enabling open-source training of Cross-Layer Transcoders (CLT-Forge).
- Co-lead the lab's interpretability wing with Dr. Jin, co-chairing weekly meetings, interviewing candidates, and mentoring two research projects, including a three-student study of whether Cross-Layer Transcoders should be trained on dataset-specific data.

Research Collaborator | University of Tübingen | Brown University

Remote

Advised by Dr. Ritambhara Singh and Dr. Carsten Eickhoff

Feb 2024 – Feb 2026

- Analysed the effectiveness of a novel semantic minimal-pair and symmetric text-replacement corruption scheme against Gaussian-noise-based causal mediation analysis, through activation patching and knockouts.
- Evaluated the multi-step math reasoning capability of the LLaVA-1.5, LLaVA-NeXT, LLaVA-OneVision, Qwen2-VL, Molmo, InternVL, LLaMA-3.2 and Math-PUMA vision-language models.

Research Intern | Inria, University of Lille

Remote

Advised by Dr. Debabrota Basu

Dec 2023 – May 2026

- Formalised sequential credit lending as a performative Markov decision process, in which the deployed policy shapes both applicants' wealth dynamics and the population that applies next.
- Built Eutopia, the first simulator for long-term lending in a performative environment, and showed that long-term equity requires performative learning and outcome-based fairness rather than parity in individual decisions.

Research Collaborator | Carnegie Mellon University

Remote

Advised by Rohan Pandey, Aryaman Arora and Dr. Paul Pu Liang

Apr 2023 – Dec 2023

- Proposed and developed a pipeline adapting causal mediation analysis, using Gaussian-noise injection and activation patching, to interpret black-box vision-language transformers.
- Demonstrated the method on BLIP with the COCO-QA dataset, studying the layer-wise effect of the multimodal text encoder on model outputs and how patching effects vary with noise strength.

Research Intern | University of Maryland, Baltimore

Remote

Advised by Dr. Manas Gaur

Nov 2022 – Sep 2024

- Trained general and domain-specific models for suicide risk assessment with cross-entropy and gambler's loss, on annotated social media posts classified into six wellness dimensions.
- Used singular value decomposition to analyse the impact of the loss function on the models' attention scores.

INDUSTRY EXPERIENCE

Machine Learning Intern | JP Morgan Chase

Bangalore, India

Model Risk Governance and Review Division

May 2025 – Jul 2025

- Investigated the internal mechanisms of a LLaMA-3.1 model fine-tuned on proprietary data using causal mediation analysis and activation patching; reviewed quantitative models through backtesting, SHAP and partial dependence plots.

SERVICE AND ACTIVITIES

Reviewing: ICLR 2025, NeurIPS Mechanistic Interpretability Workshop 2025, ICLR 2026, NeurIPS 2026, ICLR 2027

Mentoring: Guided over 25 students on research directions, career planning and academic development.

Volunteering: Machine Learning Summer School (MLSS) 2026, Tübingen.

Technical Writing: Authored blogs reviewing papers on ML, DL and AI.

[Medium]

TECHNICAL SKILLS

Programming Languages: Python, C/C++, MATLAB

ML-DL: PyTorch, TensorFlow, Torchvision, scikit-learn, Caffe

CV-NLP: Transformers, OpenCV, PIL, LlamaIndex

Miscellaneous: MySQL, LaTeX, HTML, Markdown, Git

AWARDS AND ACHIEVEMENTS

Master's Thesis: Highest master's thesis score in the department, with the top grade in both components.

Adobe AI Challenge: Captained the IIT Kharagpur team to Gold among 23 IITs at the Inter IIT Tech Meet 2024.

JEE Advanced 2021: Top 1% nationally.